Perception Is How We Experience: Sensory Perceiver as a Typed Multimodal Architecture with Replay-Centric Continual Learning

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Abstract

Sensory Perceiver is a multimodal architecture that combines modality-specific towers with shared cross-modal attention in settings where systems must keep learning over time. The implementation keeps modality identity explicit through runtime-validated MIME-style contracts, while the current empirical core is a replay-centric continual-learning program built on the attention and latent-bottleneck lineage running from Transformers to Perceivers [Vaswani et al., 2017, Jaegle et al., 2021, 2022]. On Split-CIFAR-100, feature-only replay fails with severe forgetting (CIL 4.86%, BWT -53.80%), raw-input replay with reservoir sampling materially improves retention (CIL 19.54%, BWT -34.91%), and the current tuned-core configuration improves the multiseed mean further to CIL 23.96% and BWT -30.39% over six Titan runs. The strongest mechanistic conclusion is therefore that replay signals must reach shared feature extractors, not only late task heads.

The strongest completed perception-side evidence is the hardened synthetic sensor-routing ladder, where typed canonical routing outperforms merged routing on the baseline, capacity-matched control, leave-one-out branches, corruption branch, and two of three single-modality noise branches, with `noise_robot_arm` as the main exception. Real-data evidence is now reported conservatively with an explicit bootstrap-versus-canonical split: bootstrap runs on Speech Commands and WISDM are includable now, `real-audio-seq-003` is the first promotable canonical integrated audio sequence, and `real-sensor-seq-002` is now the promotable canonical integrated sensor sequence. The current contribution is therefore a disciplined blueprint for building domain-specific perception systems: keep modality identity explicit, validate routing semantics at the runtime boundary, and make continual-learning safeguards act on the shared perceptual backbone.

1 Introduction

Sensory Perceiver is motivated by a simple systems question: how should a perception model be structured if it must ingest heterogeneous sensory inputs, keep modality identity explicit, and continue adapting without full retraining each time new data arrives? Attention-based sequence models made it practical to reason globally over flexible input sets [Vaswani et al., 2017], while the Perceiver family showed that a shared latent bottleneck could extend that flexibility to large, heterogeneous perceptual inputs [Jaegle et al., 2021, 2022]. The repository's answer combines three ideas:

1. modality-specific towers preserve sensor identity rather than collapsing heterogeneous streams into one generic input;
2. a shared cross-modal core integrates those streams into a common latent representation;
3. replay-centric continual learning attempts to preserve that shared representation over time.

The broad framing is machine perception, but the evidence is intentionally bounded. The strongest validated result remains the continual-learning lane on Split-CIFAR-100. The strongest completed perception-side result is the synthetic sensor-routing ladder. Real-data support exists as bootstrap evidence on Speech Commands and WISDM, and the first canonical integrated audio and sensor sequences are now promotable because their bounded reports completed

unblocked. This manuscript follows that evidence boundary rather than the broadest architectural ambition.

2 Related Work

Sensory Perceiver sits at the intersection of multimodal integration, continual learning, and systems-oriented runtime design. The architectural lineage runs through attention-based sequence models [Vaswani et al., 2017, Bahdanau et al., 2015, Hochreiter and Schmidhuber, 1997, LeCun et al., 1998, Bengio et al., 1994] and the Perceiver family [Jaegle et al., 2021, 2022], which demonstrated that a shared latent core can process heterogeneous inputs without a separate end-to-end architecture per modality. The continual-learning lineage runs through replay, reservoir sampling, EWC, SI, and gradient-projection methods, all of which address catastrophic forgetting from different directions [Kirkpatrick et al., 2017, Lopez-Paz and Ranzato, 2017, Chaudhry et al., 2019, Buzzega et al., 2020].

The implementation here differs in emphasis. The central claim is not that MIME-style modality semantics are parsed directly inside model forward passes, but that explicit modality contracts at the runtime boundary reduce interface drift and make typed routing experimentally tractable. The empirical center of gravity is replay-path design: raw-input replay through the shared backbone matters much more in the current codebase than mild follow-on regularization. That systems emphasis places the work closer to deployable perception stacks than to another incremental attention variant.

3 Methods

3.1 Architecture

Each modality is routed through its own tower and then integrated through a shared cross-modal attention core. This design follows the Perceiver intuition that large heterogeneous inputs should be compressed into a tractable latent workspace [Jaegle et al., 2021, 2022], but it layers typed runtime contracts and bounded adaptation on top of that architectural core. The current codebase treats modality identity as a canonical runtime concept rather than an implicit property of tensor shape. MIME-style modality registries are used for validation and setup derivation at the contract boundary, while downstream model dispatch remains modality-key based.

Towers should be read as modality-specific encoders for related but semantically distinct sensory views. They are often different observations of the same world-state, user state, environment, or task context, but they should not be flattened together before the model has a chance to preserve their local structure and learn how they interact. Typical tower groupings could include camera + microphone + pose, wearable + room sensors + medical device telemetry, or robot state + environment sensors + operator signals. The general architectural claim is therefore that each tower should honor the semantics of its own modality, the shared core should learn cross-modal relationships, and replay should re-enter through the towers so those semantics do not drift during continual learning.

The current implementation should be read carefully here: these towers share one `PerceiverTower` family with modality-specific configuration, rather than three completely different neural block classes. The paper's implemented claim is therefore semantic distinction first. A more mature live deployment could specialize tower internals further for temporal wearable streams, sparse ambient context, or structured actuator telemetry, but that stronger specialization remains architectural direction rather than the present code-level claim.

Broader domain-specific tower examples and future live-system specializations are tracked outside the canonical paper lane in the repository's future concept library. They should be read as conceptual architecture patterns unless and until they are implemented and evidenced in the current codebase.

Current synthetic benchmark instance. The repository's current synthetic perception-side benchmark, `sensor_streaming`, instantiates the general tower pattern with three canonical modalities:

- `smartwatch`: a 7-feature wearable stream representing body-side signals such as heart-rate and motion-derived measurements,
- `room_env`: a 4-feature ambient stream representing environmental state such as temperature, humidity, CO_2, and light,

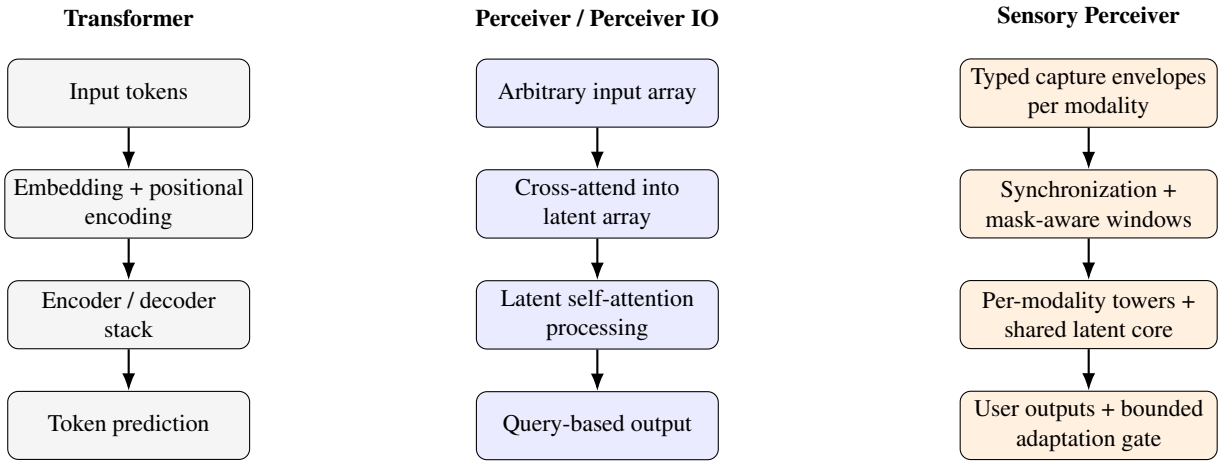


Figure 1: Positioning the system in the literature. Transformers provide the attention primitive [Vaswani et al., 2017], Perceivers provide the latent-bottleneck perception architecture [Jaegle et al., 2021, 2022], and Sensory Perceiver adds typed routing, synchronization, bounded adaptation, and replay-backed operation on top of that lineage.

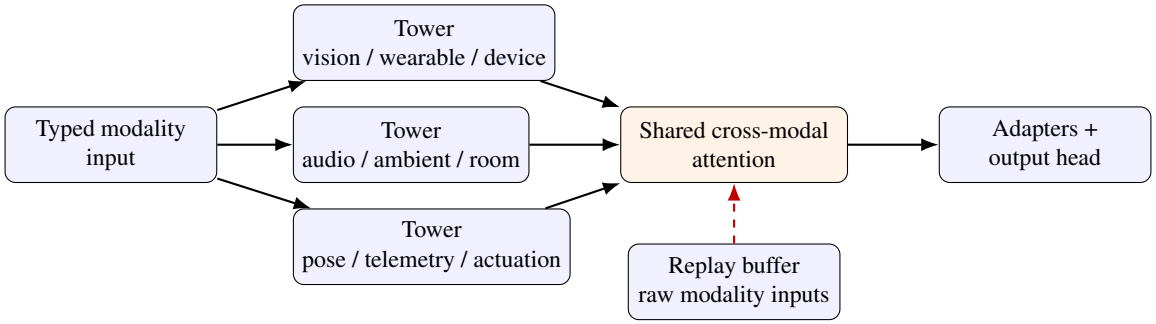


Figure 2: Sensory Perceiver keeps modality identity explicit through separate towers for semantically distinct but task-related inputs before masked shared attention learns their cross-modal relationships. The dashed replay path shows the key mechanistic result from the paper's phase progression: raw inputs must feed back into the shared backbone, not only into late heads, to improve retention. The current synthetic benchmark instance uses `smartwatch`, `room_env`, and `robot_arm` as one concrete instantiation of this more general pattern.

- `robot_arm`: a 9-feature actuation and kinematics stream representing joint and force/torque state.

Those three modalities are one benchmark instance, not the universal definition of a Sensory Perceiver deployment.

For live settings, the implementation now supports a stricter inference contract with structured per-modality payloads:

- `tensor`
- `valid_mask`
- `status`

The preferred path is native mask-aware model support, with an explicit adapter available for legacy tensor-only models.

3.2 Continual Learning

The continual-learning program is replay-centric. Phase 1 used replay too high in the stack and failed badly. Phase 2 moved replay back to raw inputs with reservoir sampling, restoring replay gradients through shared feature extractors. The current tuned-core path improves on that configuration without changing the main conclusion: replay-path design is the strongest validated retention lever in this repository. That conclusion is consistent with replay-sensitive continual-

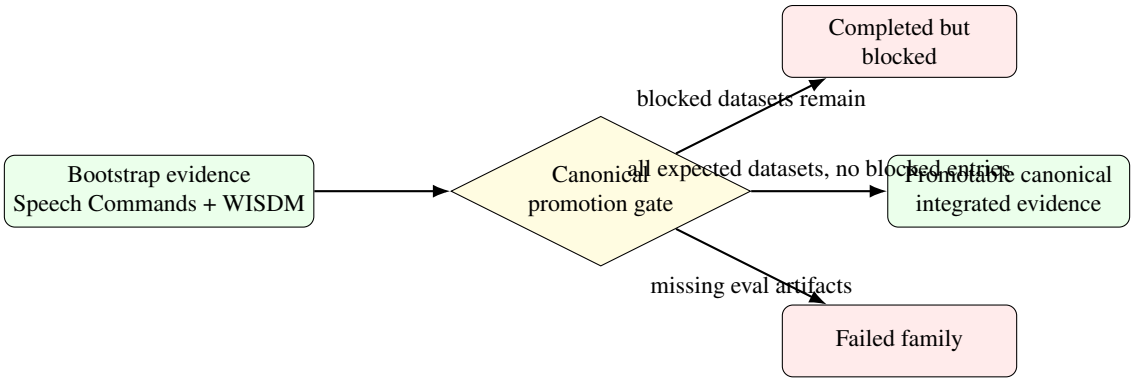


Figure 3: The bounded evidence gate used for real-data promotion. Bootstrap runs prove executability; canonical promotion requires every expected dataset family to finish successfully without blocked entries.

learning baselines in the literature, where weighting and placement of replay constraints are first-order factors rather than implementation details [Chaudhry et al., 2019, Buzzega et al., 2020].

3.3 Perception-Side Evaluation

The completed synthetic perception-side ladder compares:

- **canonical**: typed modality routing with modality-specific towers,
- **merged**: one generic sensor stream without explicit typed modality separation.

It includes a baseline, a capacity-matched control, leave-one-out branches, binding corruption, and single-modality noise stress tests. Real-data evaluation now uses an explicit evidence taxonomy:

- **bootstrap**: first executable real-data runs used to validate ingestion and end-to-end operation,
- **canonical integrated**: bounded multi-dataset sequences promoted only after successful completion.

The active bootstrap and integrated real-data sources are:

- Speech Commands
- WISDM
- ESC-50
- UrbanSound8K
- FSD50K
- RealWorld HAR via sha-ce/SensorData

4 Experiments

4.1 Continual-Learning Lane

The continual-learning lane uses Split-CIFAR-100 as the narrow but code-aligned benchmark for catastrophic forgetting. The canonical current result is the tuned-core six-seed family executed on Titan Node. The associated figures and tables are generated directly from artifact-backed runs in the results package.

4.2 Synthetic Sensor Routing Lane

The synthetic sensor ladder is complete. It is the strongest completed perception-side evidence path, but it remains synthetic and therefore bounded. Its purpose is to test whether typed routing matters under controlled stress rather than

Results Overview Table

Configuration	Source	CIL	BWT	Task 0 retention
Feature-only replay	paper_canonical_section	4.86	-53.80	0.6
aw-input replay + reservoir sampli	artifact_results_json	19.54	-34.91	33.0
ned core continual path (6-seed me	artifact_results_json_mean	23.96	-30.39	25.1

Table 1: Split-CIFAR-100 continual-learning progression. Phase 1 is a legacy feature-only replay result sourced from the canonical paper section because the original run directory is not currently tracked as a local artifact. Phase 2 and Phase 3 are artifact-backed, with Phase 3 reporting the mean over 6 tuned-core Titan runs from family `20260311_140818`.

to stand in for broad real-world multimodal validation.

4.3 Real-Data Lane

The real-data lane is intentionally conservative. Bootstrap evidence is includable now:

- `real-audio-seq-001` on `speech_commands`
- `real-seq-004` on `wisdm`

Canonical integrated audio and sensor evidence are now promotable:

- `real-audio-seq-003` on `speech_commands`, `esc50`, `urbansound8k`, and `fsd50k`
- `real-sensor-seq-002` on the integrated real-sensor lineup including `realworld_har`

Deferred legacy sources remain explicit but are no longer part of the active bounded chain:

- `audioset`: replaced operationally by `fsd50k`
- `ubiqlog`: replaced operationally by `realworld_har`

5 Results

5.1 Continual-Learning Core

The current artifact-backed continual-learning story is summarized by the generated results package:

The tuned-core six-seed result is the strongest validated configuration in the codebase:

- CIL mean: 23.96%
- BWT mean: -30.39%
- Task 0 retention mean: 25.1%

The exploratory BWT sweep remains useful negative evidence: none of the completed EWC, SI, larger-buffer ER, or contrastive replay settings surpass the tuned-core baseline.

5.2 Completed Synthetic Sensor Routing Evidence

The completed synthetic sensor ladder establishes the strongest finished perception-side result in the repository. Canonical typed routing is ahead on the baseline, the capacity-matched control, all leave-one-out branches, the

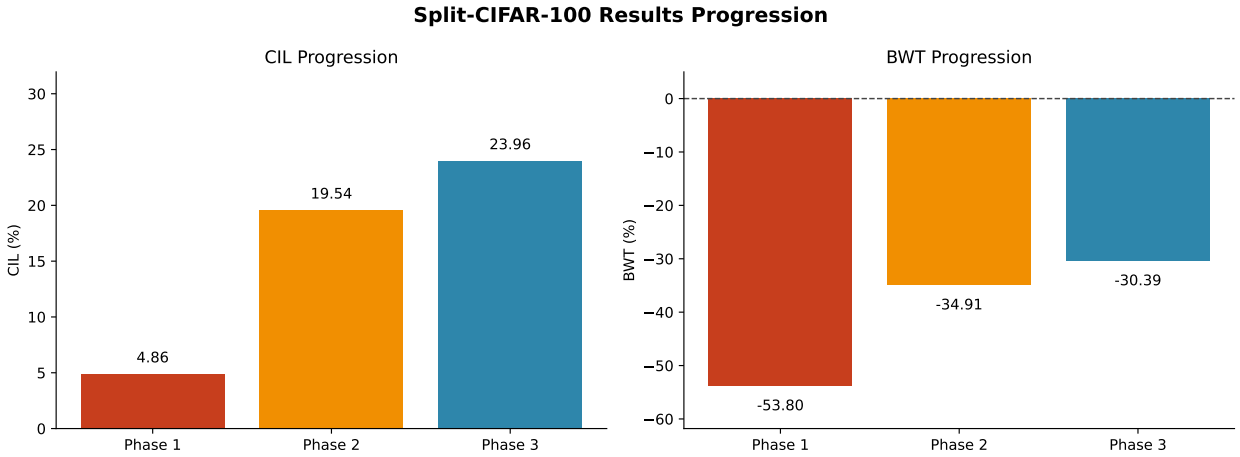


Figure 4: CIL and BWT progression from feature-only replay to raw-input replay with reservoir sampling and then the tuned-core continual configuration. The figure emphasizes that replay-path design, not just stronger regularization, drove the major retention gains.

binding-corruption branch, and two of the three single-modality noise branches. The paired gaps are:

- baseline: $+0.4583$
- capacity-matched baseline: $+0.3767$
- drop smartwatch: $+0.3317$
- drop room environment: $+0.7950$
- drop robot arm: $+0.6333$
- corrupt bindings: $+0.2717$
- noise smartwatch: $+0.0633$
- noise room environment: $+0.0617$
- noise robot arm: -0.0933

The supervisor's final state for the ladder is `canonical-dominant`. The important nuance is that the result is strong but not uniform: `noise_robot_arm` is the one completed branch where merged routing wins.

5.3 Bootstrap Real-Data Evidence

Figure 7 summarizes the current real-data package state. The completed bootstrap results are:

- `real-audio-seq-001` on Speech Commands: mean accuracy 88.32%
- `real-seq-004` on WISDM: canonical-minus-merged gap $+4.19$ percentage points

These runs establish that the first real audio and real sensor paths work end to end on Titan. `real-audio-seq-003` is now the first promotable canonical integrated audio sequence because it completed all expected datasets without blocked entries. `real-audio-seq-002` remains historical only, because it completed with blocked datasets. `real-sensor-seq-002` is now the first promotable canonical integrated sensor sequence because it completed all expected sensor datasets without blocked entries. The generated package under `paper/figures/generated/results_package` is the source of truth for bootstrap-versus-canonical status.

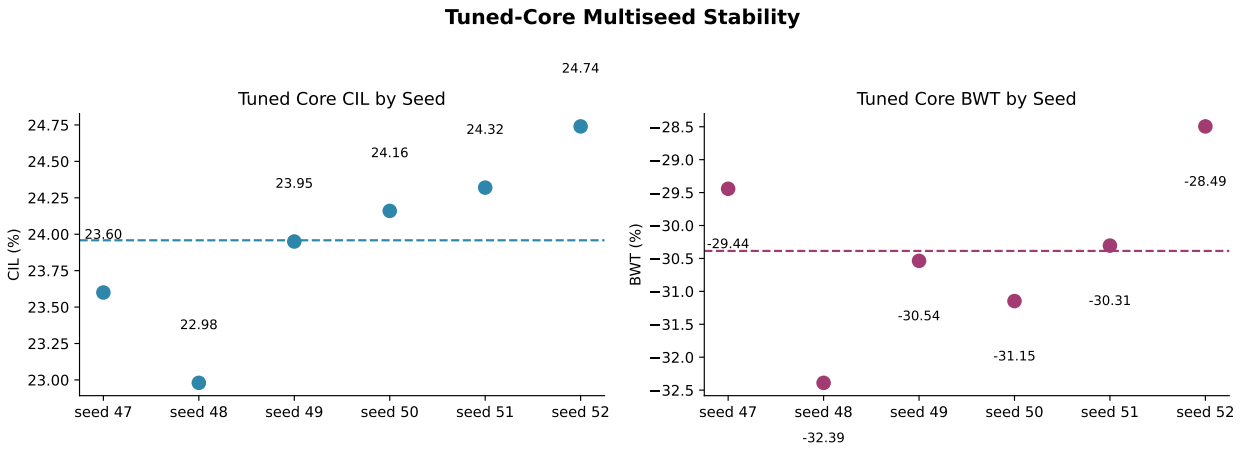


Figure 5: Tuned-core multiseed stability on Titan Node. Each point is one full 20-task Split-CIFAR-100 run, with dashed lines indicating the multiseed means. Variance is modest relative to the mean effect size.

6 Discussion

6.1 What Is Strongest Right Now

The strongest evidence in the repository is still replay-centric continual learning. The central mechanistic insight survives every implementation pass so far: replay must reach shared feature extractors, not only late task heads. The tuned-core Split-CIFAR-100 result is therefore still the main quantitative anchor for the paper.

6.2 What The Synthetic Sensor Ladder Adds

The completed synthetic sensor ladder materially strengthens the broader perception story. It shows that typed routing is not just runtime hygiene; it changes performance on a controlled benchmark, survives a capacity-matched control, and remains favorable under multiple structured stressors. At the same time, the `noise_robot_arm` reversal prevents overclaiming. The right interpretation is "strong controlled support" rather than "uniform dominance."

6.3 What The Real-Data Lane Adds

The real-data lane now matters because it prevents the project from staying trapped in synthetic-only evidence. Bootstrap Speech Commands and WISDM runs prove that the real-data path is executable. Both integrated sequences have now cleared the evidence gate: `real-audio-seq-003` completed all expected audio datasets with no blocked entries, and `real-sensor-seq-002` completed all expected sensor datasets with no blocked entries. This bootstrap-versus-canonical split remains the main guardrail against overstating progress while Titan is still resolving bounded evidence chains.

6.4 Claim Surface Versus Evidence Surface

The architectural concept is broader than the current evidence. That is acceptable only if the manuscript stays explicit about the gap. Today the evidence hierarchy is:

1. strongest validated result: replay-centric continual learning,
2. strongest completed perception-side result: synthetic sensor routing ladder,
3. completed bootstrap real-data evidence: Speech Commands and WISDM,
4. canonical integrated real-data evidence: not yet promotable.

That hierarchy is the one this manuscript defends.

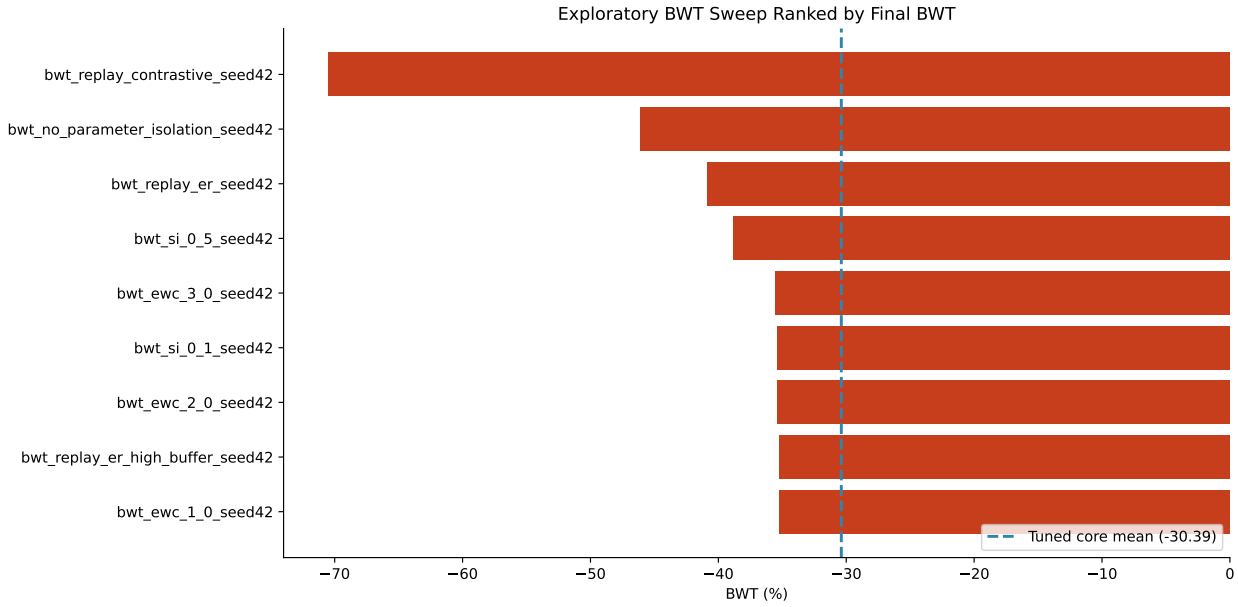


Figure 6: Exploratory BWT sweep ranked by final BWT. The dashed reference line marks the tuned-core multiseed mean. None of the completed EWC, SI, larger-buffer ER, or contrastive replay runs surpass the tuned-core baseline.

6.5 Practical Interpretation

For an applied reader, the present result is not a finished universal perception product. It is a disciplined blueprint for building domain-specific systems that:

- keep modality identity explicit,
- validate modality semantics at the runtime boundary,
- remain useful when one stream is degraded,
- and adapt over time without constant full retraining.

7 Conclusion

Sensory Perceiver now has one coherent evidence story. Replay-centric continual learning remains the strongest validated result, with the tuned-core six-seed Titan family as the current quantitative anchor. The completed synthetic sensor ladder provides the strongest finished perception-side evidence, showing a strong but not unconditional advantage for typed routing. Bootstrap real-data evidence on Speech Commands and WISDM establishes that the real-data path is executable, while canonical integrated promotion remains blocked until bounded sequences finish cleanly.

This is the right stopping point for the current manuscript. It does not overclaim incomplete evidence, it preserves deferred legacy sources such as AudioSet and UbiqLog, and it keeps the broad machine-perception framing tied to what the code and artifacts actually support. If future bounded sequences finish cleanly, the next revision can promote them into the canonical results package. Until then, the present paper should be read as a code-aligned architecture paper with a strong continual-learning core, a completed synthetic perception-side ladder, and a cautiously advancing real-data evidence path.

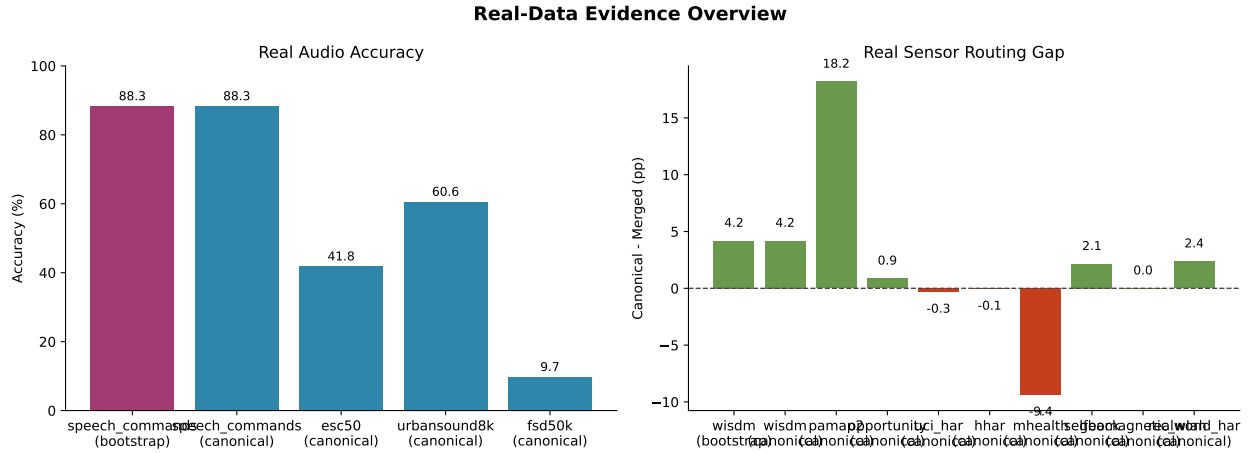


Figure 7: Real-data evidence overview. Bootstrap evidence comes from `real-audio-seq-001` on Speech Commands and `real-seq-004` on WISDM. Canonical evidence is promoted only when bounded integrated sequences finish with all expected datasets present and no blocked entries. `real-audio-seq-003` is the current canonical audio sequence, while `real-sensor-seq-002` is now promotable after completing cleanly.

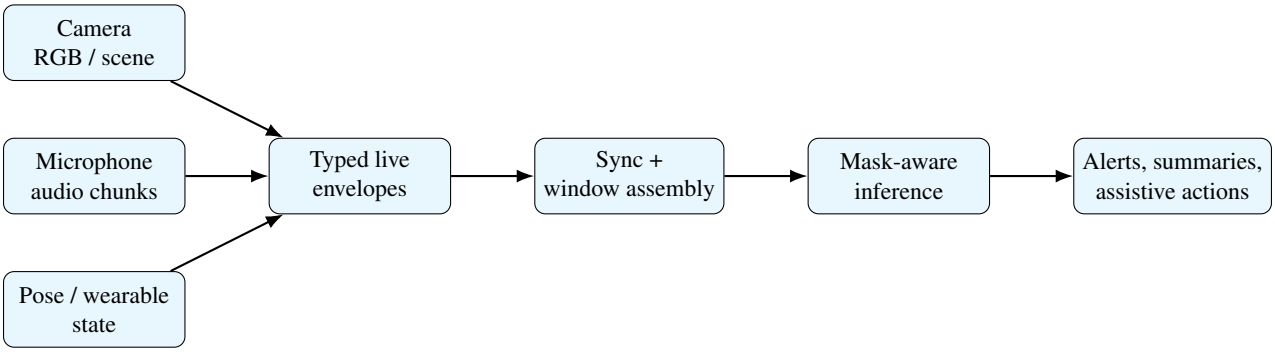


Figure 8: A concrete live deployment pattern. Multiple live sensory channels are preserved as typed envelopes, synchronized into bounded windows, then passed into mask-aware inference before being surfaced as alerts, summaries, or assistive actions.

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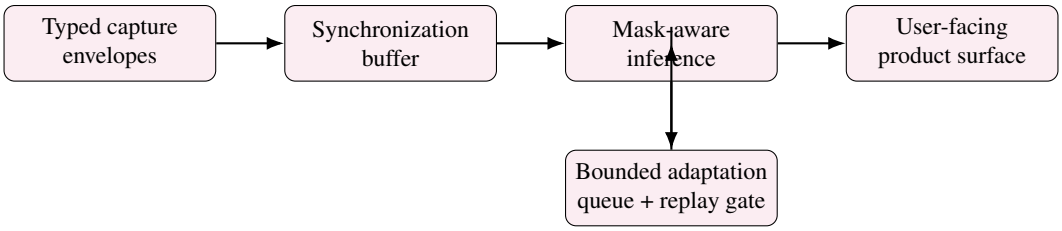


Figure 9: Live-system interpretation of the repository. Inference can be continuous, but adaptation remains bounded, replay-backed, and auditable rather than fully online and self-mutating.

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